

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – MINI APPLICATION DEVELOPMENT (CHATBOT /
SUMMARIZER / TRANSLATOR)

Course Code:	CSA65	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO4 – Precision (BL3, BL4,BL5)	Assessment:	Assessment Tool 1-Mini Application Development (Chatbot / Summarizer / Translator)
Weightage:	20% of CIA	Total Marks:	2 * 50= 100
CO4 Statement:	<i>Design and implement multimodal Generative AI applications integrating text, image, and speech models for engineering applications.(BL3, BL4,BL5)</i>		
Student Name:	N Goutham	Reg. No.:	192472024
Section / Batch:	CSE-AI	Date:	21-08-2026

Instructions to Students

- **Answer any 2 questions. Each question carries 50 marks. Total: 100 marks.**
- Implement the solution using **Python**.
- Select and justify an appropriate AI/LLM model or technique.
- Develop a **working application with a user interface**.
- Test the application using multiple input cases.
- Demonstrate handling of invalid, empty, or unexpected inputs.
- Clearly explain the application architecture and workflow.
- Submit the source code, test results, screenshots/demo, and documentation.
- Explain the limitations of the selected model/technique and suggest possible improvements.

Code of Conduct

I N Goutham/192472024 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N Goutham

SECTION A – MINI APPLICATION DEVELOPMENT (CHATBOT / SUMMARIZER / TRANSLATOR)

9. Technical Support Chatbot – Develop a chatbot that assists users with common technical problems such as password issues, software installation, login problems, network problems, and application errors. The chatbot should provide step-by-step solutions.

1. Title

Technical Support Chatbot Using Generative AI

2. Problem Statement and Objective

Users often face common technical problems such as password issues, login failures, software installation problems, network connectivity issues, and application errors.

The objective of this project is to develop an AI-based Technical Support Chatbot that accepts user queries and provides clear, relevant, and step-by-step solutions.

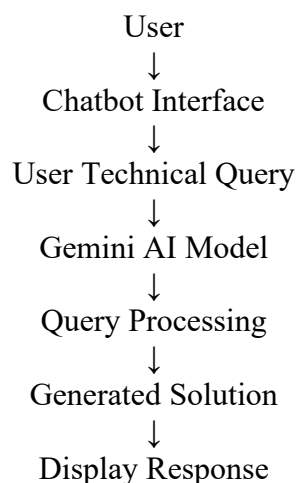
3. Technologies and AI Model Used

- **Programming/Platform:** Google AI Studio
- **AI Model:** Gemini
- **Technique:** Generative AI and Natural Language Processing

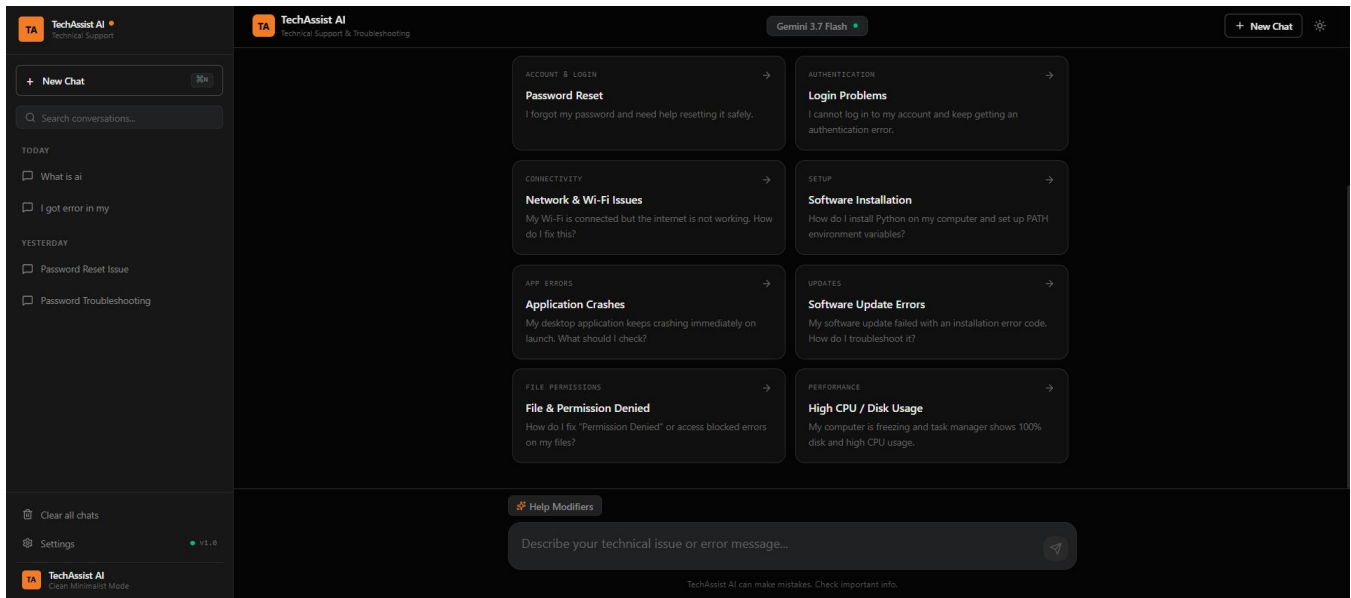
Why Gemini?

Gemini was selected because it can understand natural language queries and generate contextual responses. It can provide explanations, bullet points, and step-by-step troubleshooting solutions based on the user's technical problem.

4. Application Architecture



The user enters a technical problem through the chatbot interface. The Gemini model processes and understands the query, generates an appropriate solution, and displays the response to the user.



5. Working Process

The application works in the following steps:

1. The user enters a technical problem.
2. The chatbot receives the user's query.
3. Gemini AI analyzes the query.
4. The model identifies the technical issue.
5. Gemini generates a relevant solution.
6. The response is displayed to the user.

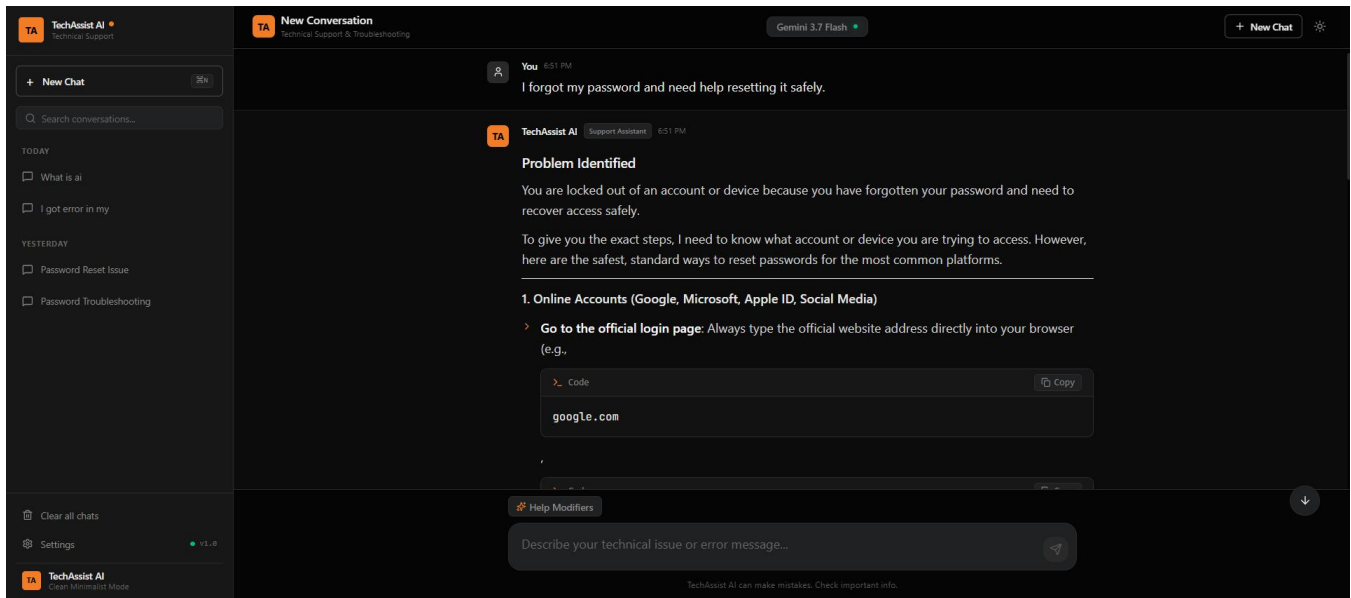
6. Application Features

The chatbot can provide support for:

- Password and account recovery issues.
- Login and authentication problems.
- Software installation guidance.
- Network and Wi-Fi connectivity issues.
- Application errors and troubleshooting.
- Step-by-step technical solutions.

7. Test Cases and Results

Test Case	Input	Result
Password Issue	I forgot my password.	Password recovery steps generated
Login Problem	I cannot log in to my account.	Login troubleshooting provided
Software Installation	How do I install Python?	Installation steps generated
Network Problem	My Wi-Fi is not working.	Network troubleshooting provided
Application Error	My application is showing an error.	Possible solutions generated



8. Invalid and Empty Input Handling

The application was also tested with empty or unclear inputs. The chatbot prompts the user to provide a valid technical question instead of processing an empty request.

Example:

“Please enter a valid technical problem or question.”

9. Limitations and Future Improvements

Limitations

- Responses may occasionally be inaccurate.
- Complex technical problems may require human support.
- The application depends on AI and internet/API availability.
- The chatbot cannot directly inspect the user's device.

Future Improvements

- Add conversation memory.
- Support multiple languages.
- Add voice input and voice responses.
- Allow users to upload screenshots of errors.
- Connect the chatbot to a technical knowledge base.

10. Conclusion

The Technical Support Chatbot was successfully developed using Generative AI. The chatbot accepts natural language queries and provides relevant solutions for common technical problems such as password issues, login problems, software installation, network issues, and application errors.

This application demonstrates how Large Language Models can be used to provide intelligent, interactive, and user-friendly technical support.

19. Article Summarizer with Key Points – Create an application that accepts a long article and generates both a paragraph summary and a list of important key points.

1. Title

Article Summarizer with Key Points Using Generative AI

2. Problem Statement and Objective

Reading long articles and identifying the important information can be time-consuming. Users may need a quick way to understand the main content without reading the complete article.

The objective of this project is to develop an AI-based Article Summarizer that accepts a long article as input and generates a concise summary along with the important key points.

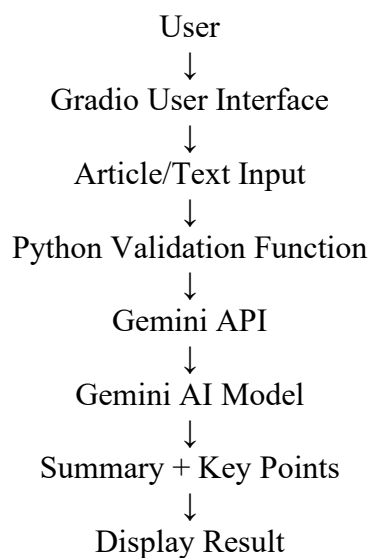
3. Technologies and AI Model Used

- **Programming Language:** Python
- **Development Platform:** Google Colab
- **User Interface:** Gradio
- **AI/LLM Model:** Gemini
- **Technique:** Generative AI and Text Summarization

Why Gemini?

Gemini was selected because it can understand the context of long text and generate meaningful summaries. It can identify the main ideas, preserve the original meaning, and extract important information in a concise format.

4. Application Architecture



The user enters an article through the Gradio interface and selects the required summary length. The application validates the input and sends the article to the Gemini AI model. Gemini processes the content and generates a concise summary along with important key points, which are displayed separately.

[Insert Screenshot of the Application/Workflow Here]

5. Working Process

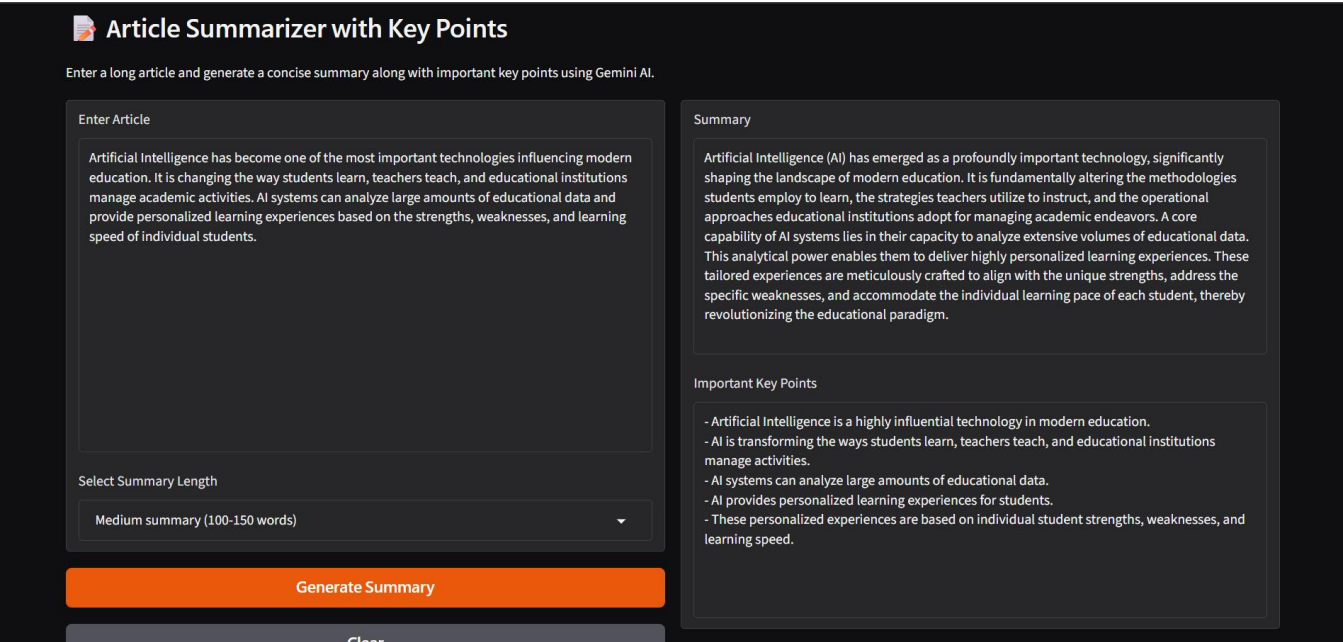
The application works as follows:

- 1. The user enters a long article or text.
- 2. The user selects the required summary length.
- 3. The application checks whether the input is valid.
- 4. The article is sent to the Gemini AI model.
- 5. Gemini analyzes the article and identifies the main content.
- 6. The system generates a summary and important key points.
- 7. The results are displayed through the Gradio interface.

6. Application Features

The main features of the application are:

- Accepts long articles or text as input.
- Provides different summary length options.
- Generates a concise AI-based summary.
- Extracts important key points.
- Handles empty input.
- Handles very short input.
- Provides a user-friendly Gradio interface.
- Includes a Clear button to reset the input and output.



7. Test Cases and Results

Test Case	Input	Result
Normal Article	Long article about Artificial Intelligence in Education	Summary and key points generated
Short Summary	Long article with short summary option	Concise summary generated

Test Case	Input	Result
Medium Summary	Long article with medium summary option	Medium-length summary generated
Detailed Summary	Long article with detailed summary option	Detailed summary generated
Empty Input	No article entered	Warning message displayed
Short Input	Less than 20 words	Validation message displayed

Article Sumarizer with Key Points

Enter a long article and generate a concise summary along with important key points using Gemini AI.

Enter Article

Paste your article here...

Select Summary Length

Medium summary (100-150 words)

Generate Summary

Clear

Summary

Please enter an article to summarize.

Important Key Points

8. Invalid and Empty Input Handling

The application validates the user input before sending it to the AI model.

For empty input, the system displays:

 Please enter an article to summarize.

For very short input, the system displays:

 Please enter a longer article with at least 20 words.

This validation helps prevent invalid requests and improves the usability of the application.

[Insert Screenshot of Empty/Invalid Input Test Here]

9. Limitations and Future Improvements

Limitations

- Important details may sometimes be omitted from the summary.
- The quality of the result depends on the input article.
- AI-generated content may occasionally contain inaccurate interpretations.
- The application requires internet access and Gemini API availability.

Future Improvements

- Add PDF and document upload support.
- Support multiple languages.
- Allow users to specify an exact summary length.
- Extract keywords and important topics.
- Add support for summarizing content from uploaded documents.

10. Conclusion

The Article Summarizer with Key Points was successfully developed using Python, Gradio, and the Gemini AI model. The application accepts a long article as input and generates a concise summary along with the most important key points.

This project demonstrates how Generative AI can help users quickly understand lengthy content by reducing it into a clear and structured format.

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Model/Technique Fit	30%	Correct, appropriate model/technique choice for the chosen application	Mostly appropriate choices made	Some inappropriate technique choices	Technique choice fundamentally flawed
Functional Completeness	25%	Application fully functional; solves a real use-case effectively	Mostly functional application	Partially functional application	Application is non-functional
Robustness & Reliability	20%	Handles edge cases; stable, reliable performance	Handles common cases well	Limited robustness to edge cases	Frequent failures during use
UI / Demo Polish	15%	Polished UI/demo with an engaging walkthrough	Clear demo, minor polish issues	Basic or rough demo	No usable demo presented
Design & Usage Documentation	10%	Comprehensive documentation of design & usage	Adequate documentation	Minimal documentation	No documentation provided

Marks Evaluation:

Question No.	Model/Technique Fit (30%)	Functional Completeness (25%)	Robustness & Reliability (20%)	UI / Demo Polish (15%)	Design & Usage Documentation(10%)	Total Marks (100)
Q1						
Q2						
Overall Total (100)						